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Triple Bottom Line-Based Dynamic System Modeling for Sustainable Rice Food Security in East Java

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Abstract: This study addresses the strategic engineering management challenge of balancing short-term farm economic profitability with long-term soil carrying capacity restoration in staple crop systems. It aims to evaluate policy trade-offs and inter-agency coordination mechanisms across the Triple Bottom Line (TBL) dimensions—People, Planet, and Profit—to enhance regional rice food security in East Java, Indonesia. A system dynamics (SD) simulation model was engineered using STELLA software, integrating demographic demand, biophysical soil degradation/regeneration loops, farmer financial investment flows, and policy execution dynamics. Model parameters were calibrated using empirical historical data (2021–2025) and stakeholder interviews ($N = 45$). Structural validity was evaluated through dimensional consistency, extreme-condition tests, and mean relative error (E1), defined as the average percentage deviation between simulated and historical behavior (E1 $\leq 4.73\%$). Four policy intervention scenarios were simulated over a 2026–2030 horizon: Baseline, Economic Intervention, Environmental Intervention, and a Combined Policy Intervention. The Baseline scenario experiences saturation after 2028, with the Sustainable Rice Food Security Index (SFSI) declining to 70.41 by 2030 due to unmitigated soil health degradation. The Economic Intervention scenario enhances immediate farmer profitability (SFSI = 75.43), but performance levels off over time due to ongoing biophysical soil depletion. The Environmental Interventio scenario restricting subsidized chemical fertilizer to 1,034,209 tons/year and transitioning to organic inputs exhibits steady ecological resilience (SFSI = 77.61). The Combined Policy Intervention scenario achieves the highest projected performance (SFSI = 81.12), buffering short-term financial transition shocks while supporting soil carrying capacity restoration. Overall, the simulation results suggest that biophysical soil health restoration constitutes a primary structural leverage point preventing limits-to-growth system degradation. Successful policy execution requires an integrated inter-agency roadmap: the National Food Agency must align price floors, the Ministry of Agriculture must reallocate chemical subsidies toward bio-organic supply chains and extension training, and the Ministry of Environment and Forestry must enforce regional soil organic monitoring.

Keywords: System dynamics; Sustainable Rice Food Security Index; Agricultural system dynamics; Rice supply chain; Fertilizer subsidy policy; Soil regeneration

1 Introduction

Rice is a critical staple crop essential for fulfilling global dietary requirements [1]. In Asian rural populations, including Indonesia, rice serves as the primary staple food contributing up to 70% of daily caloric intake, while national food security strategies also emphasize local cereal diversification to complement primary supply [2]. Consequently, ensuring a stable, sufficient rice supply serves as a primary national security benchmark [3]. However, modern agricultural management faces complex, intertwined systemic pressures [4, 5]. Expanding population growth drives continuous demand expansion [6], colliding directly with finite natural resources, environmental degradation, and shifting climate patterns [7]. Food security governance must therefore transcend traditional volume-oriented stocking strategies [8] to embrace long-term sustainability frameworks [9, 10].

In sustainable agricultural engineering management, staple crop production must be evaluated across the three Triple Bottom Line (TBL) pillars: People (social), Planet (environmental), and Profit (economic) [11]. From

the social perspective, food security relies on equitable public purchasing power and protecting smallholder farmer livelihoods across vulnerable supply chains [12, 13]. From the environmental perspective, conventional intensive farming frequently exceeds ecological carrying capacities, resulting in agricultural land conversion [14], irrigation stress [15], biophysical soil health degradation from excessive agrochemical inputs [13], and elevated carbon footprints [16]. Economically, farming enterprises must maintain financial efficiency and local economic viability [17]. Single-dimensional policy interventions inevitably create unintended negative externalities across non-targeted TBL dimensions over time.

Because cause-and-effect relationships within agricultural systems involve non-linear feedback loops, multi-year time delays, and accumulative stock dynamics, traditional linear modeling techniques are inadequate [1, 18]. Exogenous economic interventions (Profit) alter social adoption capacity (People) and induce non-linear biological soil reactions (Planet). To capture these dynamic complexities, system dynamics (SD) offers a rigorous methodology [19]. SD models explicit feedback structures, projects systemic trajectories over multi-year horizons, and enables risk-free policy scenario testing prior to real-world implementation [20].

From an engineering management standpoint, the central operational challenge facing decision-makers is resolving the policy execution conflict and inter-agency coordination barrier between short-term farm revenue protection (managed by the Ministry of Agriculture and the National Food Agency) and long-term biophysical soil carrying capacity preservation (governed by the Ministry of Environment and Forestry). While prior studies have applied SD to food security [21, 22] or evaluated livestock systems [23], comprehensive mathematical SD frameworks that couple biophysical soil carbon regeneration equations with economic TBL indices for staple crops remain sparse. Building upon our foundational multi-subsystem SD architecture established in dairy supply chains [23], this study completely re-engineers structural formulations to model soil organic matter dynamics, chemical fertilizer degradation loops, land cropping constraints, and paddy-to-rice milling conversion ratios.

The primary objective of this study is to design, validate, and simulate a TBL-based SD model to evaluate policy scenarios for sustainable rice food security in East Java, Indonesia. East Java represents an ideal regional scope as it serves as one of Indonesia's primary rice-producing provinces, characterized by high production volumes and substantial regional consumption [24], while simultaneously serving as Indonesia's principal grain basket facing intense land-use conversion pressures [14], climate vulnerability [25], and persistent soil organic matter depletion [26]. The resulting model provides regional agricultural planners with an interactive decision-support platform to optimize policy sequencing, balance subsidy allocations, and align multi-agency sustainability strategies.

2 Literature Review

The evaluation of staple food systems requires an integrated understanding of multi-dimensional interactions across social equity, environmental ecology, and supply chain economics. While traditional agricultural modeling has relied heavily on econometric regressions and static optimization techniques, these linear approaches fail to capture the feedback delays, accumulation thresholds, and non-linear dynamics inherent in tropical agro-ecosystems. Consequently, researchers have increasingly adopted SD and TBL frameworks to evaluate supply chain sustainability.

In the domain of agricultural TBL modeling, Elyasi and Teimoury [11] applied a qualitative meta-methodology to examine sustainable rice supply chains in Iran, illustrating the soft systemic interactions between stakeholder governance and environmental compliance. Similarly, Narassima et al. [17] developed a multi-staged TBL framework combined with Multi-Criteria Decision Making (MCDM) to evaluate processing efficiency and waste reduction in Asian rice mills. However, both studies focused primarily on qualitative mapping or static operational stages, lacking continuous differential equations to project multi-year biophysical soil feedback mechanisms onto crop productivity.

Addressing dynamic systemic complexity, Papanicolaou et al. [19] engineered a comprehensive SD framework to model economic, environmental, and social trade-offs in Midwestern U.S. agroecosystems undergoing transitions from row-crop agriculture to mixed farming. While their work demonstrated the efficacy of SD in evaluating soil quality and farm revenue, their structural parameters were calibrated for temperate climate regimes and large-scale mechanized farming, rendering their equations inapplicable to tropical smallholder rice systems in Southeast Asia. Table 1 presents a structured gap analysis matrix positioning this study against foundational literature in agricultural TBL and SD modeling.

In our previous research [23], we established an overarching multi-subsystem SD integration architecture to optimize livestock supply chains, validating systemic behavior using statistical error protocols (E1, E2). However, staple crop agriculture exhibits fundamentally distinct biophysical mechanisms compared to livestock systems. Livestock models center on herd demography, mortality curves, milk yield kinetics, and feed conversion ratios. In contrast, staple rice production requires explicit mathematical formulations for biophysical soil organic matter depletion, chemical fertilizer acidification penalties, biological recovery time lags, land cropping intensity constraints, and the milling yield ratio from paddy to rice. This study retains the systemic multi-subsystem coupling logic and validation protocol of Murnawan and Purnomo [23] while building entirely new structural formulations tailored to tropical crop agronomy and Southeast Asian subsidy governance.

Table 1. Positioning matrix of this study against existing agricultural Triple Bottom Line and system dynamics literature

Study	Focus Area	Methodology	Triple Bottom Line Dimensions	Policy Scenario	Main Limitation/Research Gap
[11]	Rice supply chain in Iran	Meta-methodology and soft systems	Social, Environmental, Economic	Qualitative scenario mapping	Lacks quantitative simulation and biophysical soil differential equations
[17]	Multi-stage rice supply chain	Triple Bottom Line framework and Multi-Criteria Decision Making	Economic, Environmental	Operational processing options	Focuses on downstream processing; omits dynamic feedback of soil health on crop yield
[19]	U.S. Midwestern agroecosystem	System dynamics	Social, Environmental, Economic	Row-crop-to-mixed-farming transition	Calibrated for temperate mixed farming; does not capture tropical rice milling and Asian subsidy policies
[23]	Dairy production optimization	System dynamics	Social, Environmental, Economic	Technical and feed interventions	Developed for livestock asset health; lacks crop biophysical soil carbon dynamics and land conversion logic
This study	Staple rice security in East Java	Triple Bottom Line-coupled system dynamics	Social, Environmental, Economic	Baseline, Economic Intervention, Environmental Intervention, Combined Policy Intervention	Fills the gap by mathematically coupling biophysical soil degradation loops with Triple Bottom Line policy trade-offs

3 Methodology

Building upon established multi-subsystem SD modeling procedure [23], this study combines qualitative causal modeling and quantitative stock-and-flow formulation within STELLA software to capture complex cause-and-effect dynamics [27]. The modeling process follows five structured stages: data parameterization, boundary definition, mathematical formulation, structural/behavioral validation, and scenario sensitivity analysis.

3.1 Data Collection and Parameter Calibration

Model parameters were established using empirical statistical data (2021–2025) from Statistics Indonesia (BPS) and the Regional Department of Agriculture in East Java. Empirical variables—including agricultural land area (ALA = 1,841,346 ha), birth rate (BR = 0.01320 year^{-1}), mortality rate (MR = 0.00788 year^{-1}), and baseline chemical fertilizer usage (CFU = 1,723,682 tons/year) were parameterized directly from historical records.

For qualitatively abstract parameters, primary data were collected through semi-structured stakeholder interviews conducted between October 2025 and January 2026. A purposive sampling procedure was used to select 45 domain experts across key agricultural districts in East Java (Mojokerto, Jombang, Nganjuk, and Lamongan), comprising:

- 25 lead farmers and managers from active agricultural cooperatives
- 10 regional agricultural extension officers
- 10 senior officials from the East Java Department of Agriculture and Regional Food Security Agency

Qualitative interview responses on 5-point Likert scales were transformed into normalized crisp values (V) using a linear min-max normalization:

$$V = \frac{L - L_{\min}}{L_{\max} - L_{\min}} \tag{1}$$

where,

V = normalized crisp parameter value (dimensionless, [0, 1])
L = raw expert consensus score on a 5-point Likert scale ($1 \leq L \leq 5$)
$L_{\min}$ = minimum scale value (= 1)
$L_{\max}$ = maximum scale value (= 5)

For example, an expert consensus score of $L = 4$ ("High Effectiveness") yields $V = \frac{4-1}{5-1} = 0.75$, representing a 75% operational efficacy coefficient in the STELLA model. Table 2 presents the detailed parameterization matrix, baseline values, units, sources, and calibration methods to ensure complete model reproducibility.

Table 2. Parameter calibration and baseline metadata

Parameter Name	Baseline Value	Measurement Unit	Data Year	Primary Source	Calibration/Quantification Method
Agricultural Land Area (ALA)	1,841,346	Hectares (ha)	2025	Statistics East Java	Direct empirical baseline measurement
Birth Rate (BR)	0.01320	year^{-1}	2025	Statistics East Java	Empirical annual birth rate (13.20/1,000)
Mortality Rate (MR)	0.00788	year^{-1}	2025	Statistics East Java	Empirical annual mortality rate (7.88/1,000)
Per Capita Consumption (PCC)	92.50	kg/person/year	2025	Statistics East Java	Direct empirical consumption rate
Rice Consumption Trend (RCC)	0.005	Fractional (0–1)	2025	Trend analysis	Annual shift coefficient in per capita intake
Farmers Group Participation (FGP)	0.60	Fractional (0–1)	2025–2026	Field interview	Linear min-max normalization of Likert scores
Access to Training (ATT)	0.55	Fractional (0–1)	2025–2026	Field interview	Linear min-max normalization of survey responses
Technology Adoption (TA)	0.50	Fractional (0–1)	2025–2026	Field interview	Calibrated adoption rate function
Productivity Rate (PR_{base})	5.85	tons/ha/year	2025	Statistics East Java	Baseline yield capacity multiplier
Baseline Production Cost (PC_{base})	7,150,000	IDR/ton	2025	Statistics cost survey	Fixed empirical unit cost baseline
Baseline Rice Selling Price (SP_{base})	12,000,000	IDR/ton	2025	National Food Agency	Baseline price floor for milled rice
Government Regulation (GR)	0.70	Fractional (0–1)	2025–2026	Field interview	Linear min-max normalization of regulatory index
Subsidy Intervention (SUB)	500,000	IDR/ton	2025	Ministry of Agriculture	Target annual net profit benchmark
Baseline Profit Target (PT_{base})	15,000,000	IDR/farmer	2025	Farmer survey	Fraction of profit reinvested into farm capital
Farmer Investment Fraction (IF)	0.15	Fractional (0–1)	2025	Farm financial survey	Off-farm income share proportion
Income Diversification (ID)	0.30	Fractional (0–1)	2025–2026	Field interview	Calculated non-debt equity ratio
Financial Independence (FI)	0.45	Fractional (0–1)	2025–2026	Field interview	Standard conversion ratio of paddy to rice
Paddy-to-Rice Milling Yield (η_{mill})	0.62	Fractional (0–1)	Official standard	Statistics & Ministry of Agriculture	Regional farming population divisor
Active Farming Households (N_{farmer})	2,450,000	Households	2025	Statistics East Java	Baseline organic matter quality fraction
Initial Soil Health (SH_0)	0.65	Fractional (0–1)	2025	Agronomic soil literature	Natural annual organic depletion fraction
Soil Depletion Rate (k_{dep})	0.05	year^{-1}	2025	Agronomic literature	Bio-organic restoration efficiency constant
Organic Regeneration Constant (k_{reg})	1.0×10^{-7}	tons^{-1}	2025	Model calibration	Baseline organic input volume
Organic Fertilizer Usage (OFU_{base})	350,000	tons/year	2025	Department of Agriculture	Baseline organic input volume
Chemical Fertilizer Usage (CFU_{base})	1,723,682	tons/year	2025	Ministry of Agriculture	Baseline chemical input allocation
Climate Impact (CI)	0.05	Fractional (0–1)	2025	Meteorology, Climatology, and Geophysical Agency/Regional data	Environmental yield loss penalty factor
Soil Organic Matter Depletion (SOMD)	0.12	Fractional (0–1)	2025	Agronomic data	Baseline chemical acidification index

3.2 System Definition and Boundaries

The operational boundaries encompass the complete rice production value chain in East Java over a 5-year simulation horizon (2026–2030) with a numerical integration step (Δt = 0.25 years). The model couples four interdependent subsystems:

• People subsystem (social): Manages human demography, birth/mortality dynamics, per capita consumption (PCC) trends, and labor participation in agricultural extension training.

• Planet subsystem (environmental): Governs soil biophysical health dynamics, chemical fertilizer acidification

penalties, organic matter restoration rates, and land productivity limits.

• Profit subsystem (economic): Models farmer enterprise revenues, unit production costs, net profit accumulation, organic investment capability, and financial independence.

• Policy and Sustainable Rice Food Security Index (SFSI) subsystem: Integrates normalized outputs across TBL dimensions into the composite SFSI.

3.3 Qualitative Modeling (Causal Loop Diagram)

A causal loop diagram (CLD) is used to map the feedback structure of a sustainable rice food security system based on TBL aspects. Standard conventions for constructing a CLD [27–29] include: 1) arrows indicate the direction of the relationship from cause (upstream) to effect (downstream); 2) a positive symbol (+) denotes a relationship where variables move in the same direction (as the source variable increases, the target variable increases); 3) a negative symbol (−) denotes a relationship where variables move in opposite directions (as the source variable increases, the target variable decreases). Figure 1 illustrates the CLD mapping the systemic feedback structure.

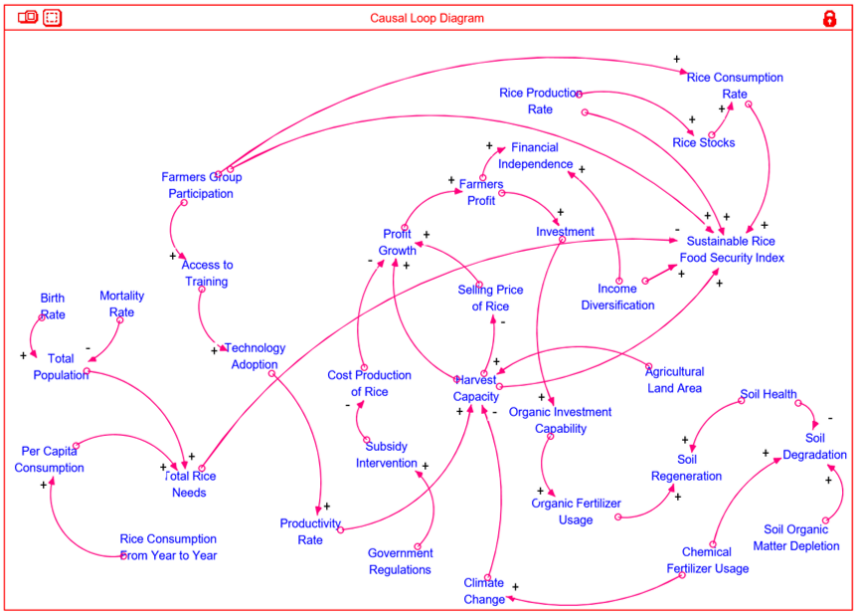


Figure 1. Causal loop diagram of the Triple Bottom Line-based sustainable rice food security model

System behavior is governed by three primary feedback loops:

• Balancing economic loop (B1): Increased harvest capacity expands rice supply, exerting downward market pressure on selling prices, which moderates profit growth and prevents excessive market overproduction.

• Reinforcing soil degradation loop (R1): Heavy CFU accelerates biophysical soil organic matter depletion (soil degradation). Decreasing soil health reduces crop yields, driving farmers to apply additional chemical fertilizers to meet short-term targets, further degrading long-term soil health.

• Reinforcing ecological recovery loop (R2): Elevated farmer profits enable higher reinvestment in organic fertilizers (organic investment capability). Organic applications accelerate soil regeneration, restoring soil health, boosting yield capacity naturally, and strengthening future enterprise profits.

3.4 Quantitative Modeling (Stock and Flow Diagram)

The stock and flow diagram (SFD) represents a formal stage for transforming the CLD conceptual model into a quantitative model incorporating mathematical formulations [27, 30]. Within the social aspect of the SD model, the primary stock variable is the human population; this is driven by BRs and MRs, which in turn influence PCC and total rice demand. This leads to the emergence of a specific stock within the economic aspect—namely, farmer profit—which is driven by the rate of profit growth and the level of investment, thereby influencing financial self-sufficiency. Finally, regarding the environmental aspect, the primary stock is soil health, which is determined by the rates of soil regeneration and degradation. Figure 2 illustrates the quantitative SFD layout.

Key structural equations governing state accumulations and flow rates are formulated as follows:

$$\begin{aligned} \frac{d(\text{Population})}{dt} &= (BR - MR) \times \text{Population} \\ \text{Total Rice Demand} &= \text{Population} \times PCC \times (1 + RCC) \end{aligned} \tag{2}$$

where, BR is the birth rate (year^{-1}), MR is the mortality rate (year^{-1}), PCC is per capita consumption (kg/person/year), and RCC is the rice consumption trend coefficient.

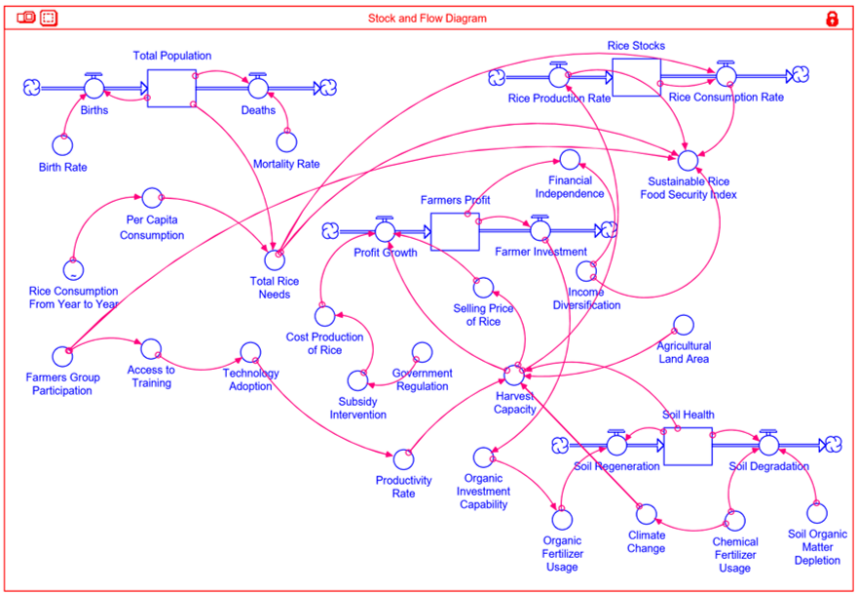


Figure 2. Stock and flow diagram of the Triple Bottom Line-based sustainable rice food security model

To explicitly represent the economic subsystem and profit growth dynamics, enterprise net profit accumulation (Π) is driven by total revenue minus production costs, adjusted by price mechanisms and subsidies:

$$\frac{d(\Pi)}{dt} = \text{Gross Revenue} - \text{Total Production Cost} \tag{3}$$

where, gross revenue is determined by harvested yield multiplied by market selling price (PS_{base}), and total production cost is governed by baseline unit cost (PC_{base}) and fertilizer inputs. A defined fraction of accumulated profit is reinvested into farm capital via the investment fraction (IF):

$$\text{Reinvestment Capital} = \Pi \times IF \tag{4}$$

For the environmental subsystem, soil health accumulation is governed by organic regeneration (Regen) minus chemical degradation:

$$\frac{d(SH)}{dt} = \text{Regen} - (k_{\text{dep}} \times SH) \tag{5}$$

where, Regen is modeled as:

$$\text{Regen} = k_{\text{reg}} \times OFU \tag{6}$$

where:
$\frac{d(SH)}{dt}$ = rate of change in soil health (year^{-1})
k_{reg} = organic regeneration constant (tons^{-1})
OFU = organic fertilizer usage (tons/year)
SH = soil health index (dimensionless, [0, 1])
Soil depletion rate (k_{dep}) is driven by CFU, soil organic matter depletion (SOMD), and climate stressors (CI):

$$k_{\text{dep}} = f(CFU, SOMD, CI) \tag{7}$$

where:
CFU = chemical fertilizer usage (tons/year)
k_{dep} = soil depletion rate (year^{-1})
$SOMD$ = soil organic matter depletion index (fractional, [0, 1])

Total agricultural production and supply-demand balancing are formulated as:

$$\begin{aligned} \text{Total Production} &= ALA \times PR_{\text{base}} \times SH \times (1 - CI) \\ \frac{\text{Deficit}}{\text{Surplus}} &= (\text{Total Production} \times \eta_{\text{mill}}) - \text{Total Rice Demand} \end{aligned} \tag{8}$$

where:

ALA = agricultural land area (ha)
PR_{base} = productivity rate (tons/ha/year)
SH = soil health (dimensionless, [0, 1])
CI = climate impact factor (fractional, [0, 1])
η_{mill} = paddy-to-rice milling yield ratio (fractional, [0, 1])

Financial independence (FI) evolves based on net savings ratio and debt equity adjustments:

$$\frac{d(FI)}{dt} = \frac{\Pi - PT_{\text{base}}}{PT_{\text{base}}} \times (1 - IF) \tag{9}$$

where:

IF = investment fraction (fractional, [0, 1])
PT_{base} = baseline profit target (IDR/farmer/year)
Π = enterprise net accumulation (IDR/farmer/year)

To fully represent the Triple Bottom Line within a single metric, the composite SFSI is formulated by integrating standardized sub-indices across Social (I_{soc}), Economic (I_{econ}), and Environmental (I_{env}) dimensions:

$$SFSI = (w_1 \times I_{\text{soc}}) + (w_2 \times I_{\text{econ}}) + (w_3 \times I_{\text{env}}) \tag{10}$$

where, the sub-indices are defined as:

$$I_{\text{soc}} = \left(\frac{FGP + ATT}{2} \right) \times 100 \tag{11}$$

where:

FGP = farmer group participation (fractional, [0, 1])
ATT = access to training (fractional, [0, 1])

$$I_{\text{econ}} = \left(\frac{\Pi}{PT_{\text{base}}} \times (1 + ID) \right) \tag{12}$$

where:

PT_{base} = baseline profit target (IDR/farmer/year)
ID = income diversification (fractional, [0, 1])

$$I_{\text{env}} = (SH \times 100) \tag{13}$$

The composite weighting structure assigns w_1 = 0.35 (35% Social), w_2 = 0.35 (35% Economic), and w_3 = 0.30 (30% Environmental), representing an author-defined analytical scenario baseline. This weighting scheme adapts the balanced multi-criteria weighting approach of the National Food Security Index framework and aligns with expert survey consensus scores (N = 45) across agricultural districts in East Java.

To maintain numerical stability and dimensional integrity, each raw composite output is restricted to a normalized range of [0, 100] using an explicit min-max clamping function:

$$SFSI = \max(0, \min(100, SFSI)) \tag{14}$$

Under this boundary formulation, potential out-of-bound simulation values falling below 0 (e.g., severe chemical degradation penalties) are clamped at 0, whereas values exceeding 100 (e.g., exceptional profit growth periods) are capped at 100, ensuring that the aggregate SFSI remains mathematically bounded within [0, 100] across all multi-year policy scenarios.

3.5 Model Validation

Model validation combined structural checking with empirical behavioral metrics across historical series (2021–2025). Structural integrity was verified via STELLA unit-consistency audits, boundary adequacy reviews, and extreme-condition stress testing (e.g., simulating zero chemical input versus 10× baseline input, yielding non-negative, non-divergent trajectories). Behavioral validity was evaluated using mean relative error (E1 $\leq$ 5%) and variance relative error (E2 $\leq$ 30%) [23, 31]:

$$E1 = \left| \frac{\bar{S} - \bar{A}}{\bar{A}} \right| \times 100\% \tag{15}$$

$$E2 = \left| \frac{S_S - S_A}{S_A} \right| \times 100\% \tag{16}$$

where,

$\bar{S}$ and $\bar{A}$ represent simulation and empirical means, respectively.

S_S and S_A denote simulation and empirical standard deviations across five annual observations.

Annual percentage errors ($APE_t = \left| \frac{S_t - A_t}{A_t} \right| \times 100\%$) were calculated to ensure low mean errors did not conceal single-year spikes.

3.6 Scenario Development

Four policy scenarios were evaluated over the 2026–2030 simulation horizon:

• Scenario 1 (Baseline): Continuation of existing agricultural trends without policy changes (CFU = 1,723,682 tons/year, Price = IDR 12,000,000/ton, Subsidy = IDR 500,000/ton).

• Scenario 2 (Economic Intervention): Implementation of updated price floors under National Food Agency Regulation No. 04/2024 [32], combined with an author-assumed effective unit subsidy proxy of IDR 640,000/Ton. This proxy value serves as an assumed operational parameter for policy simulation—derived by dividing estimated regional budget allocations by physical quotas under Ministry of Agriculture Regulation No. 01/2024 [33], rather than an officially prescribed regulatory subsidy rate.

• Scenario 3 (Environmental Intervention): An author-defined assumption capping chemical fertilizer use at a 40% restriction level (CFU = 1,034,209 tons/year), supported by empirical findings from Han et al. [34] regarding partial chemical fertilizer substitution with bio-organic inputs to optimize yield and reduce greenhouse gas emissions.

• Scenario 4 (Combined Policy Intervention): Simultaneous implementation of economic price supports (Scenario 2) and environmental organic transitions (Scenario 3).

4 Results

4.1 Structural and Behavioral Model Validation Results

Table 3 presents the empirical validation results comparing model trajectories against observed historical data from BPS and regional authorities (2021–2025) across five key system variables.

As presented in Table 3, the dynamic state variables generally satisfy both behavioral validation thresholds (E1 $\leq$ 5% and E2 $\leq$ 30%), confirming that the STELLA model accurately replicates historical system trajectories across key agricultural subsystems in East Java.

However, an exception occurs for chemical fertilizer usage, where the calculated variance relative error (E2 = 100%) exceeds the standard validation threshold (E2 $\leq$ 30%). This mathematical outcome arises because the baseline simulation models chemical fertilizer input as a constant value (CFU = 1,723,682 tons/year), resulting in a simulation standard deviation of zero (S_S = 0). While the exceptionally low mean relative error (E1 = 0.04%) demonstrates high baseline accuracy in capturing average historical consumption, we explicitly acknowledge the limitation of assuming a constant simulated input, which understates actual historical year-to-year input fluctuations [23, 31].

4.2 Policy Scenario Simulation Trajectories (2026–2030)

Simulation trajectories for the composite SFSI across all four scenarios are summarized in Table 4 and illustrated in Figure 3.

Under Scenario 1 (Baseline), SFSI rises initially from 70.27 in 2026 to a peak of 70.64 in 2028. However, after 2028, the index experiences a continuous downward decline, falling to 70.41 by 2030. The simulation indicates that unmitigated chemical fertilizer usage induces biophysical soil degradation (soil health declines from 0.65 to 0.58), creating a systemic limits-to-growth ceiling.

Under Scenario 2 (Economic Intervention), price supports and fertilizer subsidies elevate SFSI to 75.34 in 2026, reaching a peak of 75.62 in 2028, with a 7.13% increase relative to the Baseline scenario in 2030. However, the

trajectory mirrors the baseline decay pattern after 2028, declining to 75.43 by 2030. Purely economic incentives fail to arrest physical soil health depletion.

Under Scenario 3 (Environmental Intervention), restricting chemical fertilizers and mandating organic transitions generates continuous, resilient growth (the SFSI increases from 77.03 in 2026 to 77.61 in 2030, representing a 10.23% increase relative to the Baseline scenario in 2030). Controlling agrochemical inputs breaks the reinforcing soil degradation loop (R1), enabling natural biophysical recovery.

Under Scenario 4 (Combined Policy Intervention), integrating price guarantees with organic transitions achieves the highest score (the SFSI reaches 81.12 in 2030, representing a 15.21% increase relative to the Baseline scenario). Scenario 4 combines immediate economic revenue protection with long-term ecological restoration.

Table 3. Multi-subsystem behavioral validation results (2021–2025)

Year	Total Population (People)	Rice Stocks (tons)	Harvest Capacity (tons)	Chemical Fertilizer Usage (tons)	Annual Farmer Profit (Million IDR/Farmer/year)
	Actual | Model (APE)	Actual | Model (APE)	Actual | Model (APE)	Actual | Model (APE)	Actual | Model (APE)
2021	40,897,100 | 42,335,444 (3.52%)	3,552,705 | 3,845,552 (8.24%)	5,652,705 | 5,838,922 (3.29%)	1,750,000 | 1,723,682 (1.50%)	12.5 | 12.1 (3.20%)
2022	41,149,900 | 42,560,668 (3.43%)	3,893,330 | 4,261,735 (9.46%)	5,500,802 | 5,615,228 (2.08%)	1,735,000 | 1,723,682 (0.65%)	12.8 | 12.4 (3.13%)
2023	41,416,407 | 42,787,091 (3.31%)	4,313,330 | 4,526,606 (4.95%)	5,607,132 | 5,617,615 (0.19%)	1,710,000 | 1,723,682 (0.80%)	12.2 | 12.6 (3.28%)
2024	41,812,300 | 43,014,718 (2.87%)	4,650,753 | 4,766,916 (2.50%)	5,353,519 | 5,750,371 (7.41%)	1,700,000 | 1,723,682 (1.39%)	13.1 | 12.9 (1.53%)
2025	42,089,300 | 43,243,557 (2.74%)	5,027,320 | 5,051,649 (0.48%)	5,627,320 | 5,840,660 (3.79%)	1,720,000 | 1,723,682 (0.21%)	13.5 | 13.2 (2.22%)
Mean	41,473,001 | 42,788,296	4,287,488 | 4,490,492	5,548,295 | 5,732,559	1,723,000 | 1,723,682	12.82 | 12.64
$E1 < 5\%$	3.17%	4.73%	3.32%	0.04%	1.40%
$E2 < 30\%$	25.72%	20.88%	9.05%	100.00%	2.31%

Note: APE = absolute percentage error; E1 = mean relative error (average percentage deviation between simulated and historical values); E2 = variance relative error (discrepancy ratio between simulated and historical standard deviations); CFU = chemical fertilizer usage. For CFU, because the baseline simulation input is held constant (CFU = 1,723,682 tons/year, standard deviation = 0), the calculated variance relative error yields E2 = 100% which exceeds the standard E2 $\leq$ 30% threshold. Nevertheless, the exceptionally low mean relative error (E1 = 0.04%) supports its baseline accuracy for policy scenario projections.

Table 4. Simulated Sustainable Rice Food Security Index trajectories (2026–2030)

Scenario	2026	2027	2028	2029	2030	Change vs. Baseline (2030)
Scenario 1 (Baseline)	70.27	70.57	70.64	70.51	70.41	–
Scenario 2 (Economic Intervention)	75.34	75.57	75.62	75.53	75.43	+7.13%
Scenario 3 (Environmental Intervention)	77.03	77.37	77.45	77.57	77.61	+10.23%
Scenario 4 (Combined Policy Intervention)	79.80	80.42	80.75	80.98	81.12	+15.21%

Note: – = no change relative to the baseline scenario (0%).

4.3 Sensitivity Analysis

To test model robustness against parametric uncertainty, a one-at-a-time (OAT) sensitivity analysis was executed by sweeping key parameters (±20%), including the soil organic matter depletion rate (k_{dep}), organic regeneration constant (k_{reg}), and SFSI indicator weights (w_{env}). The complete parameter variations, structural rank invariance, and robustness evaluation results are summarized in Table 5.

As presented in Table 5, across all 16 sensitivity iterations, the relative performance ranking remained strictly invariant (Scenario 4 $>$ Scenario 3 $>$ Scenario 2 $>$ Scenario 1). This stability suggests that biophysical soil health recovery serves as a structurally robust leverage point, maintaining its superiority regardless of parametric variance in biophysical rates or indicator weighting schemes.

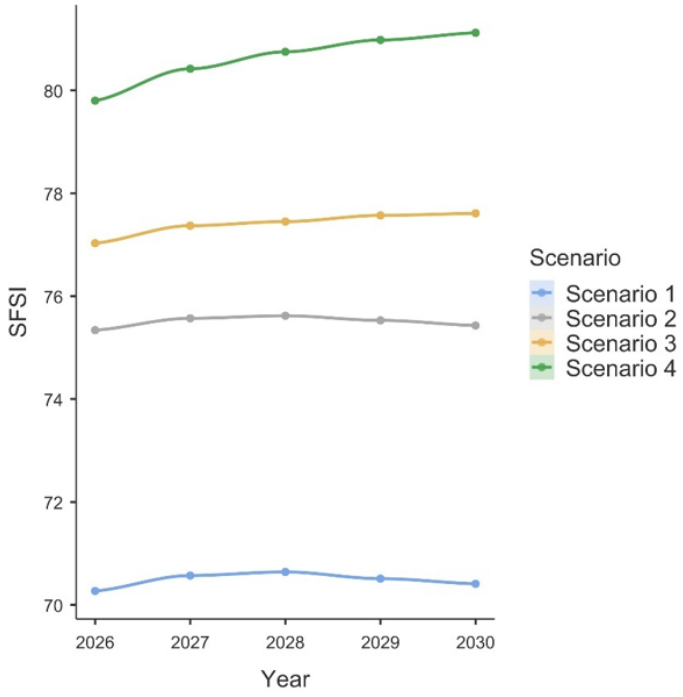


Figure 3. Simulation trajectories of the Sustainable Rice Food Security Index under scenarios 1–4

Note: SFSI = Sustainable Rice Food Security Index.

Table 5. One-at-a-time sensitivity analysis results (±20% parameter sweep)

Parameter Tested	Symbol	Baseline Value	Variation Range (±20%)	Sustainable Rice Food Security Index Rank Order (2030)	Status
Soil depletion rate	k_{dep}	0.05	[0.04–0.06]	Scenario 4 > Scenario 3 > Scenario 2 > Scenario 1	Robust
Regeneration constant	k_{reg}	1.0×10^{-7}	$[0.8 \times 10^{-7}$–$1.2 \times 10^{-7}]$	Scenario 4 > Scenario 3 > Scenario 2 > Scenario 1	Robust
Sustainable Rice Food Security Index (Planet weight)	w_{env}	0.30	[0.24–0.36]	Scenario 4 > Scenario 3 > Scenario 2 > Scenario 1	Robust
Milling yield factor	η_{mill}	0.62	[0.55–0.68]	Scenario 4 > Scenario 3 > Scenario 2 > Scenario 1	Robust

Note: All tests suggest policy rank stability across 16 simulation iterations.

5 Discussion

The dynamic simulation results support the central working hypothesis of this study: evaluating staple crop food security solely through short-term economic protection (Profit) or volume availability (People) induces systemic limits-to-growth ceilings, whereas biophysical soil organic restoration (Planet) serves as a primary structural leverage point for long-term supply chain resilience.

5.1 Inter-Agency Decision Trade-offs and Hypothesis Verification

The comparative trajectories across Scenarios 1 to 4 provide actionable insights into operational trade-offs facing regional food governance. In evaluating pure economic stimulus (Scenario 2), upstream price supports managed by the National Food Agency under Regulation No. 04/2024 [32], alongside input subsidies allocated under Ministry of Agriculture Regulation No. 01/2024 [33], cogenerate an immediate financial buffer for farming households. However, the post-2028 trajectory flattening, during which the SFSI drops from 75.62 to 75.43 indicates that monetary stimulus alone cannot fully offset underlying physical soil degradation. Building on prior agroecosystem SD research [35], this study demonstrates that economic protectionism without ecological safeguards yields diminishing returns over multi-year horizons.

Conversely, biophysical soil restoration acts as a systemic leverage point, as observed in the Environmental Intervention (Scenario 3). Phasing chemical fertilizer caps alongside organic substitution, in alignment with decarbonization strategies and optimized bio-organic application frameworks [34], effectively breaks the reinforcing

soil degradation loop (R1). Improving soil health restores crop yields naturally, thereby aligning with sustainable agriculture frameworks [10, 11], which emphasize that ecological carrying capacity forms an essential foundation for social accessibility and long-term farm income.

Resolving these inter-agency conflicts requires an integrated execution approach, as modeled in Scenario 4. The combined policy results indicate that economic incentives and environmental caps are not mutually exclusive strategies. By pairing minimum guaranteed purchase prices [32] with subsidized bio-organic fertilizer distribution and partial nitrogen substitution [34], decision-makers achieve SFSI Performance (= 81.12 by 2030). This integrated configuration addresses historical policy frictions between revenue-focused regulatory frameworks and long-term conservation mandates.

5.2 Implementation Feasibility and Transition Management

Translating the projected performance of Scenarios 3 and 4 into real-world execution requires managing critical operational friction points that lie outside pure differential equation models. Primary among these friction points is the biological time lag inherent in soil recovery, where biophysical soil organic matter regeneration typically requires a continuous 2–3-year application cycle. An abrupt 40% reduction in chemical fertilizers without sufficient organic transition buffers risks inducing short-term yield shocks and temporary farm income losses. Furthermore, scaling bio-organic inputs across 1.84 million hectares in East Java presents supply chain constraints, suggesting that regional authorities should establish localized composting facilities and bio-fertilizer processing centers to guarantee timely input delivery.

Beyond physical supply chain bottlenecks, farmer behavioral resistance and extension capacity represent notable social hurdles; smallholder farmers frequently resist reducing agrochemical inputs due to perceived yield risks. Overcoming this behavioral inertia requires expanding agricultural extension services and participatory farmer field schools to demonstrate organic efficacy. Finally, enforcing chemical fertilizer distribution quotas and verifying organic soil compliance introduces administrative monitoring overhead for regional supervisory commissions.

To systematically mitigate these transition risks, regional planners should adopt a phased three-stage execution roadmap. Phase 1 (years 1–2) focuses on pilot district demonstration and infrastructure development across major rice-producing centers (e.g., Mojokerto and Nganjuk), while expanding extension training programs, constructing regional organic processing units, and providing temporary financial cushions to buffer farmers during the transition. Subsequently, Phase 2 (years 3–4) enforces progressive chemical fertilizer quota reductions of 10–15% annually, strictly synchronized with the scaling and distribution of subsidized bio-organic fertilizers. Ultimately, Phase 3 (year 5 onward) achieves full regional policy integration across East Java, sustained by real-time digital soil health monitoring networks to support long-term ecological and economic stability.

5.3 Agency-Specific Action Matrix and Boundary Limitations

To ensure practical utility for decision-makers in engineering management, the operational responsibilities across regional food governance entities are structured around explicit inter-agency action mandates. Under this coordinated framework, the National Food Agency is responsible for conditioning minimum guaranteed purchase prices on farmer compliance with sustainable soil management practices [32]. Simultaneously, the Ministry of Agriculture must reallocate existing chemical fertilizer subsidy budgets toward scaling bio-organic supply chains and expanding agricultural extension training [33]. To complement these economic and technical measures, agricultural extension services and regional authorities are mandated to establish soil organic carbon monitoring networks, drive mandatory organic fertilizer substitution, and enforce broader agricultural decarbonization targets [34].

Despite these actionable insights, the validity and generalizability of these scenario projections remain strictly bounded by the structural scope of this study, as SD models are inherently tailored to their defined boundary conditions [27]. Geographically and temporally, the simulation model is calibrated exclusively for the regional rice supply chain of East Java, Indonesia, spanning a 2026–2030 simulation horizon. Furthermore, because the model focuses intentionally on upstream agricultural production and soil biophysical dynamics, it explicitly excludes downstream retail price volatility, interregional distribution logistics, extreme climate disruption shocks (such as severe El Niño events), and household-level micro-economics. Consequently, these quantitative findings should not be interpreted as direct empirical evidence for all agricultural regions in Indonesia or for other staple crops without prior recalibration of local biophysical, land-use, and demographic parameters.

6 Conclusions

This study engineered and evaluated a TBL-based SD model to address rice food security governance in East Java, Indonesia. The simulation findings suggest that biophysical soil health restoration constitutes a critical leverage point required to mitigate limits-to-growth saturation in regional staple crop production. From an academic perspective, this research contributes to agricultural SD modeling by offering a quantitative framework that couples biophysical soil degradation formulations with TBL economic and social metrics.

Operationally, the results translate into a proposed agency-specific action matrix for regional food governance: the National Food Agency needs to align minimum guaranteed purchase prices with sustainable farming compliance; the Ministry of Agriculture should consider reallocating chemical fertilizer subsidies toward bio-organic supply chains and extension training; and the Ministry of Environment and Forestry is encouraged to enforce regional soil organic carbon monitoring alongside agricultural decarbonization targets.

However, the applicability of these scenario projections remains strictly bounded by the study's structural scope, which focuses on East Java's upstream production dynamics through a 2026–2030 simulation horizon. Consequently, these model-based findings should not be directly generalized to agricultural regions with contrasting biophysical or logistical profiles without recalibrating local parameters. To expand upon these boundary conditions, future research should integrate downstream retail logistics, price volatility, and extreme climate disruption shocks.

Author Contributions

Conceptualization, H.M. and H.P.; methodology, E.K. and W.N.C.; software, H.M.; validation, H.P., E.K., and W.N.C.; formal analysis, E.K.; investigation, W.N.C.; resources, H.M.; data curation, H.M. and H.P.; writing—original draft preparation, H.M.; writing—review and editing, H.M. and H.P.; visualization, E.K.; supervision, H.P.; funding acquisition, H.M. All authors have read and agreed to the published version of the manuscript.

Informed Consent Statement

Informed consent was obtained from the participants involved in the validation process, and all data were analyzed anonymously.

Ethical Approval

Ethical approval was waived for this study because it primarily utilized publicly available secondary data, and the subsequent confirmation process did not involve any personal or sensitive human intervention.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

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Conflicts of Interest

The authors declare no conflicts of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

During the preparation of this manuscript, the authors utilized Google Gemini strictly as an advanced language editing tool for English grammatical refinement, sentence restructuring, and enhancing academic phraseology. The authors declare that no generative AI tools were used to create, alter, or synthesize primary research data, system dynamics model equations, STELLA simulations, or empirical findings. All mathematical formulations, model calibrations, parameter tables, statistical validation checks (E1, E2), and policy scenario interpretations were conceived, engineered, and verified independently by the authors.

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